



A Cross-Layer Optimization and Design approach under QoS constraints for green IP over WDM networks



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ABSTRACT

Energy consumption of communication networks is growing very fast due to the rapidly increasing traffic demands. It is important and valuable to find a way to save power for such networking systems. In this paper, we propose a Cross-Layer Optimization and Design (CLOD) approach to improve the energy efficiency of Internet Protocol (IP) over Wavelength Division Multiplexing (WDM) backbone networks under QoS constraints. CLOD makes use of the skew spatial distribution (i.e. 80–20 law) of traffic demands to construct the virtual topology of IP layer with a consideration of the cross-layer resource constraints. Also, CLOD takes advantage of the constraint-based routing to satisfy the QoS constraints in terms of allowed maximum hop count and allowed maximum link utilization. Taking into account the connection between base network design and network operation, CLOD dimensions the base network by using pattern of network operation and optimizes network power consumption by reconfiguring network to adapt traffic variation and the available resources. Simulation results indicate that CLOD can save power significantly and achieve near energy-proportional networks. In addition, an analysis on trade-off performance between the hop count and power consumption is given.

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1. Introduction

In recent years, the energy consumption of communication networks has been increasing very fast. It was reported that the annual growth rate was about 10% in the interval 2007–2011 [1], and this trend will most likely continue because network traffic is still growing every year. As a result, a series of issues related to energy cost, heat dissipation and carbon footprint emerges. Reduction of energy consumption is becoming a major concern in communication networks.

Communication networks are typically over-dimensioned to handle maximum traffic demands, which causes network resources to be underutilized most of time. On the other hand, nowadays the power consumption of network equipment is almost independent of traffic load [2,3]. As a consequence, current communication networks are energy inefficient severely. Energy-proportional computing as a new computing paradigm has been proposed in [4], which requires energy consumption in proportion to the amount of work performed. This can also be applied to communication networks to require power consumption of networks in proportion to the amount of traffic load without degrading Quality of Service (QoS).

Energy can be saved by improving the utilization of network resources and putting idle network resources into sleep. The utilization of network resources can be improved

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by dividing a day into several traffic stationary periods and reconfiguring (or optimizing) network resources for each period since the traffic matrices of backbone networks exhibit strong diurnal patterns and are typically stationary with a relative long period (1 to 1.5 h) [5]. This work focus on how to reconfigure network resources for each traffic stationary period in IP (Internet Protocol) over WDM (Wavelength Division Multiplexing) networks to reduce energy consumption without degenerating QoS. This process is called in this paper as power optimization in network operation phase. The design of base network is closely related with network operation since base network is designed according to the estimated maximum traffic matrix and the using pattern of network resources in network operation phase. However, the reconfiguration of network resources in network operation phase will change the using pattern of network resources, and the maximum traffic matrix will not be enough for dimensioning the base network if a different resource using pattern is used during network operation. So, the power optimization problem in network operation phase have to be considered together with the base network design problem.

This paper first gives the formulation descriptions of these two problems, and then proposes a novel heuristic algorithm CLOD (Cross-Layer Optimization and Design) to solve these two problems consistently. The spatial distribution feature of backbone traffic demands is exploited to design the virtual topology of IP layer. Traffic grooming, the process of grouping many small traffic flows into a larger one, is used in CLOD to improve the utilization of network resources. Indeed, CLOD also takes account of QoS by controlling the maximum hop count and the maximum link utilization. Besides, single-path routing is used to avoid the jitter caused by fractional routing. Last but not least, an adaptive lightpath establishing approach is proposed to guarantee the resource constraints, which can make the resource using pattern in network operation phase adapt to the base network. Simulation results indicate that CLOD can reduce network power consumption significantly and achieve near energy-proportional IP over WDM networks under QoS constraints. Our technical contributions can be summaries as follows.

- (a) The deviation of resource using pattern in base network design and power optimization is considered first time in this work to the best of our knowledge. An effective and heuristic solution CLOD for resolving these problems is proposed.
- (b) QoS constraints are considered in CLOD which limits the maximum hop count to control the length of paths and makes sure the traffic of link does not exceed its allowed maximum bandwidth.
- (c) CLOD takes the resource constraints into account and makes the traffic routing and virtual topology (or lightpaths) actively adapt to the base network.
- (d) A comprehensive simulation study is carried out, which helps to understand the role of traffic grooming in power optimization and the impact of hop count on network power consumption.

The remainder of this paper is organized as follows. Section 2 reviews some related work on this topic; Section 3 formulates the problems to be solved in this work, that is, the base network design and the power optimization; Section 4 details our proposed CLOD; Section 5 presents performance evaluation, where the simulation settings, results and analyses are reported; and Section 6 concludes this paper with some remarks.

2. Related work

Numerous literatures have focused on energy saving in communication networks in recent years. Device- or component-level energy saving, such as [6–9], focused on reducing the energy consumption of single device or component, which can enable more energy-proportional network equipment. Another amount of work concentrated on the network-wide energy saving, which can improve the energy-proportional feature of the whole network. This work belongs to the latter, so we mainly introduce the work of network-wide energy saving, a more comprehensive survey of green networking research can be found in [10].

Network-wide energy saving usually can be formulated as an optimization problem (e.g. MILP problem [11–20] and ILP problem [21–24]) with the objective of minimizing network power consumption. The network power consumption considered by different work differed usually, but the main energy-intensive components include links (or interfaces), line cards, transceivers, regenerators, optical amplifiers (EDFA) etc. Since IP networks is much power-hungry than the underlying WDM networks, many efforts, such as [11–14,25,26], are focus on saving energy of IP networks. However some work, such as [16,27], also concentrated on reduce power consumption of WDM networks. Moreover, cross-layer energy saving researches, such as [15,17–20,23,24], takes into the power consumption of multiple network layers simultaneously. The optimization spaces, usually different for different work, include traffic routing, topology design (or reconfiguration), wavelength assignment and regenerator placement etc.

This work falls into the cross-layer energy saving research in IP over WDM networks. With respect to existing work, the differences of this study include:

- (a) Existing work such as [17–20,23] paid equal attention on the power consumption of IP layer and WDM layer which makes the optimization problem very complicated. In fact, IP layer is much power-hungry than WDM layer, which takes up to 90% power consumption of the whole IP over WDM network [17]. So this work simplifies the optimization problem by only considering the power consumption of IP layer but without losing the cross-layer constraints.
- (b) Our power model is different from the existing work. We consider the modular structure of IP routers and model their power consumption as the sum of power consumption of router interfaces, line cards and chassis and the dynamic power consumption which is dependent with the traffic load.

- (c) Existing work [15,17–20,23] using CPLEX to solve the NP-hard MILP/ILP problems with exponential complexity. This work proposed the CLOD heuristic algorithms which can obtain solutions close to optimal with polynomial complexity.
- (d) Existing heuristics [14,21,22,26] all based on the unreasonable assumption that all the lightpath requests in the virtual topology of IP layer can be accommodated by the WDM layer. This study removes this assumption and takes account of the power optimization problem in network operation phase together with the base network design problem. The CLOD algorithm proposed in this work solves the two problems consistently, which can guarantee all resource constraints between IP layer and WDM layer and between the optimized network and the base network.
- (e) This work also considers the network performance by limiting the maximum hop count to control the path length and the maximum link utilization to control the network congestion. The relationship between the hop count and network power consumption is also investigated, which can be used to decide the optimal hop count limit.

3. Problem formulation

In order to improve the energy efficiency of the IP over WDM networks under QoS constraints, two steps will be taken in our proposed solution. First, we design the IP over WDM base network according to the maximum traffic demands of operation phase. And the output in this step includes the resource provisioning (i.e. the installation of fibers and all necessary hardware at each node) and the corresponding resource using pattern (i.e. the traffic routing of both the two layers and virtual topology of IP layer). The base network can be considered as a static system since its resource using pattern is independent of demand fluctuations over time. Second, we optimize the power consumption of the base network via reconfiguring network resources (i.e. changing resource using pattern) for each traffic stationary period without degenerating QoS. In this step, the given traffic demands is the maximum traffic demands of each traffic stationary period, and the output is the resource using pattern. After the power optimization, the resource using pattern is adaptively changed with the variation of traffic demands. Table 1 gives a comparison of the two steps from four aspects. Note that the traffic demands in the two steps are different.

The QoS considered include delay, packet loss and jitter. It is well known that the end-to-end delay correlates to the

hop count and link utilization. Thus, in this paper we limit the maximum hop count and link utilization through adding corresponding constraints. In order to alleviate jitter caused by fractional routing, this work uses a single-path routing in both IP layer and WDM layer. Furthermore, packet loss is also considered by adding link capacity constraints to guarantee that the traffic of links do not exceed their maximum allowed bandwidth.

3.1. Design of the base network

Let N be the set of node of IP over WDM networks, and each node in N is composed of a node in IP layer (i.e. IP routers) and a node in WDM layer respectively. Let K denotes the set of types of chassis of IP routers. 0–1 variable $x(i, k) = 1$ if node $i \in N$ install chassis of type $k \in K$; otherwise $x(i, k) = 0$. At each node $i \in N$, $m_c(i, k)$ chassis of type $k \in K$ are installed to constitute the multi-chassis routing system, each chassis holds $m_l(k)$ line cards, and at most $\psi(k)$ chassis are allowed at a node (i.e. $m_c(i, k) \leq \psi(k)$). Each line card holds m_i interfaces of capacity c Gb/s each, and the maximum allowed utilization of each interface is μ_{max} . The number of active chassis, line cards and interfaces at each node $i \in N$ are $n_c(i)$, $n_l(i)$, and $n_i(i)$ respectively. The static power (i.e. traffic-independent power) of each chassis, line card and interface are $p_c(k)$ where $k \in K$, p_l and p_i respectively, while the dynamic power (i.e. traffic-dependent power) for a unit (i.e. 1Gbps) of traffic is p_v .

The WDM layer is modeled as a directed graph $G = (N, E)$, where E denotes the set of directed physical links (i.e., fibers) in WDM layer. Each physical link $e \in E$ is composed of $b_f(e)$ fibers at cost $\gamma(e)$, and each fiber consists of w wavelengths of capacity c Gb/s each. The set of candidate paths in WDM layer from node $i \in N$ to node $j \in N$ is $P_{wdm}(i, j)$, and if path $p \in P_{wdm}(i, j)$ include link $e \in E$, $\delta_{wdm}(p, e) = 1$, otherwise $\delta_{wdm}(p, e) = 0$. Routing algorithm in WDM layer must select a path p from the set $P_{wdm}(i, j)$ for lightpath request from node i to node j , so if path p is selected then $y_{wdm}(p) = 1$; otherwise $y_{wdm}(p) = 0$.

The IP layer employs a virtual topology in which each logic link is made up of several lightpaths provided by WDM layer. The task of virtual topology design is to determine the number of lightpaths $b_w(i, j)$ for each logic link (i, j) where $(i, j) \in N \times N$. The set of candidate paths in IP layer from node i to node j with h hop counts is $P_{ip}(i, j, h)$, and if path $p \in P_{ip}(i, j, h)$ include logic link $(u, v) \in N \times N$, $\delta_{ip}(p, u, v) = 1$, otherwise $\delta_{ip}(p, u, v) = 0$. The maximum allowed hop count in IP layer is h_m , so the inequality $1 \leq h \leq h_m$ holds. Let D be the set of traffic demands of IP layer, $s(l)$, $t(l)$ and $d(l)$ represent the source node, destination node and the value in Gb/s of traffic demand $l \in D$ respectively. Routing algorithm in IP layer must select a path p from all the candidate path sets whose hop count h is not greater than h_m (i.e. $\bigcup_{h=1}^{h_m} P_{ip}(s(l), t(l), h)$) for traffic demand $l \in D$, so if path p is selected then $y_{ip}(p) = 1$; otherwise $y_{ip}(p) = 0$. The traffic volume of each logic link $(i, j) \in N \times N$ is $t(i, j)$.

The main purpose of design of the base network is to determine the type and the number of chassis at each node, and the number of fiber for each physical link (i.e.

Table 1
Base network design vs. power optimization.

	Base network design	Power optimization
Timing	Design phase	Operation phase
Objective	Minimizing CapEx	Minimizing power
Given	Traffic demands, topology	Traffic demands, topology, resource provisioning
Output	Resource provisioning, resource using pattern	Resource using pattern

$x(i, k)$, $m_c(i, k)$ and $b_f(e)$ where $i \in N$, $k \in K$ and $e \in E$ such that there are enough network resources to accommodate the maximum traffic demands with minimal network cost. It is reported in [19] that the power consumption of the network designed with the objective of power minimization is almost identical to the power consumed by the CapEx minimized network because, nowadays in general, a network component with high CapEx normally also

consumes a lot of power. So it is reasonable in some extent to use the power consumption as the cost of IP layer. This problem can be formulated as the following MINLP (Mixed-Integer Nonlinear Programming) problem, and the notations used in the formulations are summarized in Table 2–5.

OPT-BND:

$$\begin{aligned} \text{Min} \sum_{i \in N} \left[\sum_{k \in K} x(i, k) \cdot m_c(i, k) \cdot p_c(k) \right] &+ \sum_{(i, j) \in N \times N: i \neq j} t(i, j) \cdot p_t \\ &+ \sum_{e \in E} \gamma(e) \cdot b_f(e) \end{aligned} \quad (1)$$

Subject to:

$$\sum_{h=1}^{h_m} \sum_{p \in P_{ip}(s(l), t(l), h)} y_{ip}(p) = 1 \quad \forall l \in D \quad (2)$$

$$t(i, j) = \sum_{l \in D} d(l) \cdot \sum_{h=1}^{h_m} \sum_{p \in P_{ip}(s(l), t(l), h)} y_{ip}(p) \cdot \delta_{ip}(p, i, j) \quad \forall i, j \in N : i \neq j \quad (3)$$

$$t(i, j) \leq \mu_m \cdot c \cdot b_w(i, j) \quad \forall i, j \in N : i \neq j \quad (4)$$

$$\sum_{j \in N - \{i\}} b_w(i, j) \leq n_i(i) \quad \forall i \in N \quad (5)$$

$$\sum_{j \in N - \{i\}} b_w(j, i) \leq n_i(i) \quad \forall i \in N \quad (6)$$

$$n_i(i) \leq n_l(i) \cdot m_i \quad \forall i \in N \quad (7)$$

$$\sum_{k \in K} x(i, k) = 1 \quad \forall i \in N \quad (8)$$

$$n_l(i) \leq \sum_{k \in K} x(i, k) \cdot m_c(i, k) \cdot m_l(k) \quad \forall i \in N \quad (9)$$

$$m_c(i, k) \leq \psi(k) \quad \forall i \in N, k \in K \quad (10)$$

$$\sum_{p \in P_{wdm}(i, j)} y_{wdm}(p) = 1 \quad \forall i, j \in N : i \neq j \quad (11)$$

$$\sum_{(i, j) \in N \times N: i \neq j} b_w(i, j) \cdot \sum_{p \in P_{wdm}(i, j)} y_{wdm}(p) \cdot \delta_{wdm}(p, e) \leq w \cdot b_f(e) \quad \forall e \in E \quad (12)$$

Table 2

Summary of the input parameters common for both the two formulations.

Notation	Description
$G = (N, E)$	Directed graph of the WDM layer topology where N is the set of nodes and E represents the set of physical links
D	Set of traffic demands of IP layer
$s(l)$	Source node of the traffic demand $l \in D$
$t(l)$	Destination node of the traffic demand $l \in D$
$d(l)$	Value in Gb/s of the traffic demand $l \in D$
$P_{wdm}(i, j)$	Set of candidate paths from node $i \in N$ to $j \in N$ in WDM layer
$P_{ip}(i, j, h)$	Set of candidate paths from node $i \in N$ to $j \in N$ with hop count h in IP layer
h_m	Maximum allowed hop count in IP layer
$\delta_{wdm}(p, e)$	1 if path $p \in P_{wdm}(i, j)$ in WDM layer includes link $e \in E$; 0, otherwise
$\delta_{ip}(p, u, v)$	1 if path $p \in P_{ip}(i, j, h)$ in IP layer includes logic link $(u, v) \in N \times N$; 0, otherwise
w	Number of wavelengths of each fiber
c	Capacity of each interface/wavelength
μ_m	Maximum allowed utilization of interfaces
p_i	Static power of each interface
m_i	Interface number of each line card
p_l	Static power of each line card
p_t	Dynamic power for a unit of traffic

Table 3

Summary of the decision variables common for both the two formulations.

Notation	Description
$b_w(i, j) \in Z_0^+$	Number of lightpaths of each logic link $(i, j) \in N \times N$ in IP layer, and Z_0^+ presents the set of nonnegative integers
$y_{ip}(p) \in \{0, 1\}$	1 if path $p \in P_{ip}(s(l), t(l), h)$ is selected in IP layer for demand $l \in D$; 0, otherwise
$y_{wdm}(p) \in \{0, 1\}$	1 if path $p \in P_{wdm}(i, j)$ is selected in WDM layer; 0, otherwise
$n_c(i) \in Z_0^+$	Number of active chassis at node $i \in N$
$n_l(i) \in Z_0^+$	Number of active line cards at node $i \in N$
$n_i(i) \in Z_0^+$	Number of active interfaces at node $i \in N$
$t(i, j) \in R_0^+$	Traffic volume of link $(i, j) \in N \times N$, and R_0^+ represents the set of nonnegative real numbers

Table 4

Summary of the input parameters specified to the base network design formulation.

Notation	Description
$\gamma(e)$	Cost of each fiber for link $e \in E$ (length dependent)
K	Set of type of chassis
$m_l(k)$	Line card number of each chassis of type $k \in K$
$\psi(k)$	Maximum allowed number of chassis of type $k \in K$
$p_c(k)$	Static power of each chassis of type $k \in K$

Table 5

Summary of the decision variables specified to the base network design formulation.

Notation	Description
$m_c(i, k) \in Z_0^+$	Number of chassis of type $k \in K$ at node $i \in N$. Note that in the multi-chassis cluster system, $m_c(i, k)$ is greater than 1
$b_f(e) \in Z_0^+$	Number of fibers of each physical link $e \in E$
$x(i, k) \in \{0, 1\}$	1 if chassis of type $k \in K$ at node $i \in N$ is used; 0, otherwise

The objective function (1) minimizes the cost of the whole IP over WDM network. The cost of IP layer is represented by the power consumption which consists of the static power of each node and the dynamic power of all the network traffic. The static power of each node is composed of chassis static power $\sum_{k \in K} x(i, k) \cdot m_c(i, k)$, line card static power $n_l(i) \cdot p_l$, and interface static power $n_i(i) \cdot p_i$ according to the modularity of IP routers. The dynamic power can be approximated with linear function of traffic volume [28], so dynamic power for the traffic volume of link (i, j) is $t(i, j) \cdot p_t$. The cost of WDM layer is represented by the cost of fibers of each physical link $\gamma(e) \cdot b_f(e)$.

The routing constraints of IP layer (i.e. (2)) select a path whose hop count does not exceed maximum allowed hop count for each traffic demand. It is worth noting that the traffic demands used for the base network design are the above-mentioned maximum traffic demands. QoS is considered here. First, single-path is used here to alleviate jitter, and then hop count is limited by introducing the maximum allowed hop count h_m to avoid delay increasing caused by overlong paths.

The traffic volume of each logic link (i, j) can be obtained according to (3) which accumulates all traffic demands whose paths include logic link (i, j) . And the link capacity constraints (i.e. (4)), as one type of important QoS constraints, can guarantee that the traffic of each logic link does not exceed the allowed maximum bandwidth, that is $\mu_m \cdot c \cdot b_w(i, j)$. Note that each logic link (i, j) consists of $b_w(i, j)$ lightpaths of capacity c each, and μ_m is the maximum allowed utilization of lightpath (or interface).

The resource constraints, i.e. (5)–(10), determine the type and number of chassis to accommodate the active interfaces and line cards. For each node $i \in N$, (5) and (6) require enough active interfaces accommodating the lightpaths starting from or ending at this node, and (7) makes sure enough active line cards to accommodate the active interfaces. Constraint (8) requires each node can only install one type of chassis, and (9) ensures there are enough chassis to hold the active line cards. Constraint (10) requires the number of chassis of each type at each node does not exceed the allowed maximum number of this type chassis.

Constraint (11) selects a path in WDM layer for the lightpath requests generated by each logic link (i, j) in IP layer. Constraint (12) ensures there are enough fibers to accommodate the lightpaths across each physical link, which also means the number of lightpaths transiting each physical link must not exceed its maximum number of wavelengths.

3.2. Power optimization in operation phase

The general idea of power optimization is to change the routing and virtual topology of IP layer to adapt the variation of traffic demands, and exploit sleeping technique to save power. Different from the base network design, the maximum traffic demands of the next traffic stationary period are used as the input of the power optimization. Besides, the type and number of chassis installed at each node and the number of fibers for each physical link, i.e. the outputs of the base network design, also become the

input parameters of the power optimization. Since the type of chassis at each node has been determined, we do not care about the type of chassis during the power optimization, thus we denote the number of chassis installed at each node $i \in N$ as $m_c(i)$, the power of each chassis at node $i \in N$ as $p_c(i)$, and the number of line cards of each chassis at node $i \in N$ as $m_l(i)$. These notations are summarized in Table 6. The outputs of power optimization include the routing (i.e. $y_{ip}(p)$) and virtual topology (i.e. $b_w(i, j)$) of IP layer, the routing of the WDM layer (i.e. $y_{wdm}(p)$), traffic volume of each link (i.e. $t(i, j)$), and the active interfaces (i.e. $n_i(i)$), line cards (i.e. $n_l(i)$) and chassis (i.e. $n_c(i)$) of each node. We can formulate this power optimization problem as the following MINLP problem.

OPT-PO:

$$\begin{aligned} \text{Min} \sum_{i \in N} [n_c(i) \cdot p_c(i) + n_l(i) \cdot p_l + n_i(i) \cdot p_i] \\ + \sum_{(i, j) \in N \times N: i \neq j} t(i, j) \cdot p_t \end{aligned} \quad (13)$$

Subject to: (2)–(7), (11), (12), and

$$n_l(i) \leq n_c(i) \cdot m_l(i) \quad \forall i \in N \quad (14)$$

$$n_c(i) \leq m_c(i) \quad \forall i \in N \quad (15)$$

Objective function (13) minimizes the power consumption of IP-over-WDM network. Note that we only take account of the power-hungry IP layer which takes up 90% power consumption of the whole IP over WDM network [17]. Constraint (14) requires enough active chassis to accommodate the active line cards, and (15) ensures the active chassis do not outnumber the chassis installed. The other constraints (i.e. (2)–(7), (11), (12)) are the same with that of the base network design except that the traffic demands in (3) are the maximum traffic demands of the next T hours which, in practice, can be predicted using the similar method with the dynamic cyclo-stationary model proposed in [5]. This paper mainly focuses on the algorithm design, a discussion about the implementation issues in real operational networks can be found in [29].

4. Algorithm description

It is well known that MINLP is NP-hard problem in general. There are at least $|N|^2 + 3 \cdot |N|$, where $|N|$ denotes the cardinality of set N , non-negative integer variables and $|P_{ip}| + |P_{wdm}| + |K| \cdot |N|$ 0–1 variables in our two MINLP formulations, where P_{wdm} and P_{ip} are the candidate path sets in WDM and IP layer, respectively. This huge number of variables makes it impossible to get a feasible solution even for a medium-sized network. In this paper,

Table 6

Summary of the input parameters specified to power optimization formulation.

Notation	Description
$m_c(i)$	Number of chassis at node $i \in N$, noting that in the multi-chassis cluster system, $m_c(i)$ is greater than 1
$m_l(i)$	Line card number of each chassis at node $i \in N$
$p_c(i)$	Power consumption of each chassis of node $i \in N$

we propose a heuristic framework, called CLOD (Cross-Layer Optimization and Design), to solve the two problems simultaneously.

4.1. General ideas

CLOD is based on the spatial distribution of traffic matrix that follows a rough 80–20 law (80% of traffic is generated by the largest 20%) of flows [30]. Thus the virtual topology can be constructed mainly according to these large traffic demands because of their relative importance. **Theorem 1** indicates that these large traffic demands should be routed via single-hop routing in order to save interfaces, and the needed lightpaths (or logic links) are established accordingly to make up the main virtual topology. And then the remaining small traffic demands can be groomed into the residual bandwidth of the virtual topology via multi-hop routing which can improve the utilization of logic links. If there is not enough residual bandwidth for some small traffic demands, additional new lightpaths (or logic links) should be established. However these new lightpaths may be underutilized because they are established just for a few small traffic demands, so a rerouting operation is performed at last which tries to remove the new established underutilized lightpaths.

Theorem 1. *If a traffic demand $l \in D$ is not less than the allowed maximum bandwidth of each lightpath $\mu_m \cdot c$, that is $d(l) \geq \mu_m \cdot c$, then the single-hop routing always requires the least interfaces.*

Proof. If the single-hop routing is used, k_s lightpaths (or pair of interfaces) are required to route the traffic demand $l \in D$ where $k_s = \lceil d(l)/(\mu_m \cdot c) \rceil$. Due to $d(l) \geq \mu_m \cdot c$, $k_s \geq 2$ holds. If the multi-hop routing is used, we assume a perfect traffic grooming is performed. That is to say, the fraction of traffic demand $l \in D$ (i.e. $d(l) - \lfloor d(l)/(\mu_m \cdot c) \rfloor \cdot \mu_m \cdot c$) can be groomed into other lightpaths. Let h be the hop count of multi-hop routing, so $h \geq 2$ holds and only $h \cdot k_m$ (where $k_m = \lfloor d(l)/(\mu_m \cdot c) \rfloor$ and $k_m + 1 \geq k_s \geq k_m \geq 1$ holds) lightpaths are required by traffic demand $l \in D$. Then we have $h \cdot k_m - k_s \geq 2 \cdot k_m - k_s \geq 2 \cdot k_m - (k_m + 1) = k_m - 1 \geq 0$, that is $k_s \leq h \cdot k_m$, which means single-hop routing require less lightpaths (or interfaces) than multi-hop routing. $\square$

All the constraints in the above formulations, namely routing constraints, link capacity constraints and resource constraints, are considered in CLOD. Firstly, CLOD employs the CBR-Dijkstra (i.e. constraint-based routing Dijkstra) algorithm [22] to guarantee the routing constraints and link capacity constraints. Secondly, the resource constraints are guaranteed by the base network design. In CLOD, the base network is designed with the maximum traffic matrix and the same resource using pattern with network operation. Lastly, for some small traffic demands, multi-hop routing may cause overlong paths which may increase the delay significantly. To solve this issue, the CBR-Dijkstra algorithm also implements a maximum hop count constraint which can limit the maximum hop count of traffic demands.

4.2. CLOD algorithm

Algorithm 1. CLOD.

Require: $N, E, D, c, \mu_m, p_i, m_i, p_l, p_t, \tau, \alpha, h_m, P_{wdm}, b_f(e)$
 $\forall e \in E, m_l(i), p_c(i), m_c(i) \forall i \in N$
Ensure: $power, L, P_{ip}, P_{wdm}, t(i, j), n_i(i), n_l(i), n_c(i) \forall i, j \in N$
1: $D_1 \leftarrow \{l : d(l) \geq \tau \cdot \mu_m \cdot c \wedge l \in D\}$
2: $D_2 \leftarrow \{l : d(l) < \tau \cdot \mu_m \cdot c \wedge l \in D\}$
3: **for each** $l \in D_1$ **do**
4: $x \leftarrow \lceil d(l)/(\mu_m \cdot c) \rceil$
5: establish x lightpaths from $s(l)$ to $t(l)$ for demand l
6: route demand l over the x lightpaths using single-hop routing
7: add logic link $(s(l), t(l))$ into the set L
8: add the path of demand l into the set P_{ip}
9: **end for**
10: $L_1 \leftarrow \emptyset$
11: **for each** demand $l \in D_2$ in descending order of demand values **do**
12: route demand l over the residual bandwidth of logic links L using the CBR-Dijkstra algorithm
13: **if** fail to route **then**
14: establish a new lightpath to route demand l
15: add the new logic link into the set L and L_1
16: **end if**
17: add the path of demand l into paths set P_{ip}
18: **end for**
19: **while** $L_1 \neq \emptyset$ **do**
20: get the logic link (i, j) with minimum utilization out of L_1
21: **if** the utilization of link (i, j) is greater than α **then**
22: break
23: **end if**
24: get all the demands which were routed across the link (i, j) as set D_3
25: **if** succeed to reroute all traffic demands in D_3 over the other links **then**
26: remove link (i, j) from L
27: **end if**
28: **end while**
29: Compute $n_i(i), n_l(i)$, and $n_c(i)$ for each node $i \in N$ according to (5)–(7), (14), and (15).
30: compute $power$ according to (13)

The pseudo code of the CLOD algorithm is shown as **Algorithm 1**. The input of **Algorithm 1** includes the threshold τ which is used to partition the traffic demands into large demands and small demands, the threshold of link utilization α which is used to decide the links over which the traffic flows need to be rerouted, the path set of lightpath requests in WDM layer P_{wdm} , and $N, E, D, b_f(e), c, \mu_m, p_i, m_i, p_l, m_l(i), p_c(i), m_c(i), p_t, h_m$. The output of **Algorithm 1**

includes the power consumption of the IP over WDM network power, the set of logic links L , paths of all traffic demands in IP layer P_{ip} , updated paths of all lightpath requests in WDM layer P_{wdm} , and $n_c(i)$, $n_l(i)$, $n_i(i)$, $t(i, j)$.

At the beginning, the large traffic demands are put into set D_1 (line 1), and the small traffic demands are put into set D_2 (line 2). The threshold τ should be 100% according to Theorem 1, but a smaller value (e.g. 80%) would be better because a relative large traffic demand (i.e. approaching the maximum allowed bandwidth) usually cannot be groomed into the residual bandwidth of other lightpaths.

From line 3 to line 9, Algorithm 1 establishes lightpaths for each traffic demands in D_1 (lines 4, 5), and route these traffic demands over the lightpaths via single-hop routing (line 6). The logic links are added into the set of logic links L in line 7 and the paths are added into the path set of IP layer P_{ip} in line 8. In line 5, the conditions to establish x lightpaths include (1) enough unused interfaces of source node $s(l)$ and destination node $t(l)$ and (2) the success of routing of the lightpath requests in WDM layer. Both conditions are guaranteed by the over-provisioned base network (we will give the method to design the base network immediately).

Since only the power consumption of power-hungry IP layer is considered, when new lightpaths need to be established in WDM layer (i.e. line 5 and 14), we can reuse the paths of base network in WDM layer (i.e. the paths in P_{wdm}) to reduce the unnecessary routing computation. Only when there is no path in P_{wdm} for the lightpath requests (this may happen in line 14), new paths need to be computed via CBR-Dijkstra routing in WDM layer. Also, the new paths should be added into P_{wdm} . This is the reason why the paths of WDM layer P_{wdm} are both the input and output of Algorithm 1.

From line 11 to line 18, Algorithm 1 grooms the traffic demands in D_2 over the residual bandwidth of the logic links in L using CBR-Dijkstra routing. The traffic demands are processed in descending order of their values as described in line 15, which can make sure as much as possible larger traffic demands can be groomed. If no path can be found for some traffic demands, new lightpaths should be established as shown in line 14. The logic link set L will be updated in line 15, and the path of each traffic demand will be added into P_{ip} in line 17. Link set L_1 is used to record the new established logic links (line 10 and 15), and each of them is composed of one lightpath because the value of traffic demand in D_2 is less than the maximum allowed bandwidth of single lightpath.

The logic links in L_1 which are established for a few small traffic demands may underutilized heavily, so from line 19 to line 28 Algorithm 1 tries to remove the logic links whose utilizations less than threshold via rerouting the traffic demands across these links over the other links using CBR-Dijkstra routing. In each run of the loop, the link with the minimum utilization are taken out of L_1 (line 20), and the traffic demands cross over this link are put into set D_3 (line 24). Only when all traffic demands in D_3 are rerouted successfully, this logic link is removed (lines 25–27).

The routing and virtual topology have been obtained so far, that is to say variable $y_{wdm}(p)$, $y_{ip}(p)$, $t(i, j)$ and $b_w(i, j)$ have been determined. In line 29, Algorithm 1 compute

the values of variable $n_i(i)$, $n_l(i)$, and $n_c(i)$ according to constraint (5)–(7), (14), and (15), that is:

$$n_i(i) = \max \left(\sum_{j \in N - \{i\}} b_w(j, i), \sum_{j \in N - \{i\}} b_w(i, j) \right) \quad \forall i \in N \quad (16)$$

$$n_l(i) = \left\lceil \frac{n_i(i)}{m_i} \right\rceil \quad \forall i \in N \quad (17)$$

$$n_c(i) = \left\lceil \frac{n_l(i)}{m_l(i) \cdot m_i} \right\rceil \quad \forall i \in N \quad (18)$$

Constraint (15) can be guaranteed by the over-provisioned base network. Finally, in line 30 the power consumption of the whole network power is computed according to (13).

Algorithm 2. Dimensioning chassis.

```

1: for each node  $i \in N$  do
2:    $p_{min} \leftarrow \infty$ 
3:   for each type  $k \in K$  do
4:      $m_c(i, k) \leftarrow \lceil n_l(i) / m_l(k) \rceil$ 
5:     if  $m_c(i, k) \leq \psi(k)$  and  $m_c(i, k) \cdot p_c(k) < p_{min}$  then
6:        $p_{min} \leftarrow m_c(i, k) \cdot p_c(k)$ 
7:        $m_c(i) \leftarrow m_c(i, k)$ 
8:        $p_c(i) \leftarrow p_c(k)$ 
9:        $m_l(i) \leftarrow m_l(k)$ 
10:    end if
11:  end for
12: end for

```

4.3. Design of base networks

The resource constraints of power optimization are guaranteed by the design of base network, so the dimensioning of the base network must meet the maximum traffic demands. The main idea of design of the base network is to running the power optimization algorithm (i.e. the CLOD algorithm) with maximum traffic demands (i.e. $D = D_{max}$) and unlimited network resources ($b_f(e) = \infty$, $m_c(i) = \infty$), and then the actual used network resources ($b_w(i, j)$, $n_l(i)$) are used to dimension the base network. According to (12), the fibers needed for each physic link e can be computed according to (19).

$$b_f(e) = \left\lceil \frac{\sum_{(i,j) \in N \times N: i \neq j} b_w(i, j) \cdot \sum_{p \in P_{wdm}(i,j)} y_{wdm}(p) \cdot \delta_{wdm}(p, e)}{W} \right\rceil \quad \forall e \in E \quad (19)$$

According to (8)–(10), $m_c(i)$ can be computed by Algorithm 2, where there are two conditions to choose the type of chassis (line 5): (1) the number of chassis does not exceed the maximum number of chassis, and (2) the power consumption of chassis is as low as possible. Once the type of chassis at node $i \in N$ is determined, the number of chassis at this node $m_c(i)$, the power consumption of each chassis $p_c(i)$, and the number of line cards of each chassis $m_l(i)$ can also be determined (lines 7, 8 and 9).

4.4. Guarantee of resource constraints

We have dimensioned the base network according to the maximum traffic demands so far. However, how can we know whether this base network can guarantee the resource constraints during the power optimization or not? It is well known that the traffic demands are fluctuant over time, so each power optimization will have different traffic demands. The power optimization is dependent upon the traffic demands, so different traffic demands result in the change of routing, virtual topology and resource reconfiguration (we call these as the using pattern of network resources). If the resource using pattern during the power optimization deviate that of the base network significantly, the resource constraints of power optimization will not be satisfied.

Algorithm 1 requests the over-provisioned base network to guarantee their right execution (i.e., line 5 and 14). Two types of network resources are required to establish lightpaths: (1) the interfaces of the end-nodes of lightpaths in IP layer, and (2) the bandwidth employed by the path of lightpath requests in WDM layer. In line 5, the interfaces are adequate because at first the base network is dimensioned to meet the maximum traffic demands and secondly both base network design and power optimization employ single-hop routing for large traffic demands (i.e., the same resource using pattern for large traffic demands). The bandwidth of WDM layer is also sufficient because the lightpath requests reuse the paths computed in the base network design.

In line 14, the simplest way is to establish a lightpath directly from the source node $s(l)$ to the destination node $t(l)$ so that the traffic demand l can be accommodated in a single-hop path. However, this does not always work because the base network is provisioned according to the corresponding resource using pattern of the maximum traffic matrix which may be different from the corresponding resource using pattern of the current traffic demands. We develop a more flexible method, i.e., Algorithm 3, to establish the lightpath and compute the path for demand l .

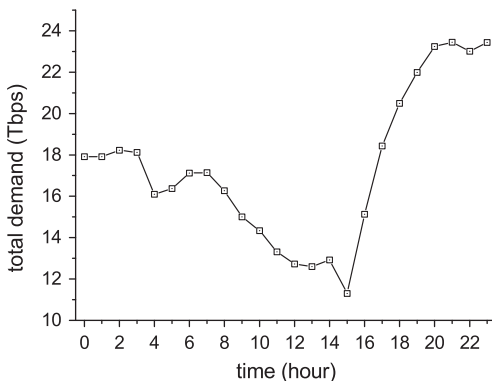


Fig. 1. The total demands of each hour's maximum traffic matrix for GEANT during a day.

Algorithm 3. Establishing lightpath

```

1: remove all edges whose residual bandwidth are
   less than  $d(l)$ 
2: put all nodes with unused interfaces and
   connected with node  $s(l)$  into set  $S$ 
3: put all nodes with unused interfaces and
   connected to node  $t(l)$  into set  $T$ 
4: compute the shortest path from node  $s(l)$  to each
   node  $i \in S$  (i.e.  $\langle s(l), \dots, i \rangle$ ) with hop count as path
   cost, and denote their hop count as  $hop(i)$ 
5: compute the shortest path from each node  $j \in T$  to
   node  $t(l)$  (i.e.  $\langle j, \dots, t(l) \rangle$ ) with hop count as path
   cost, and denote their hop count as  $hop(j)$ 
6: for  $h \leftarrow 1$  to  $h_m$  do
7:   for each  $(h_1, h_2)$  s.t.  $h_1 + h_2 = h - 1$  do
8:     for each node pair  $(i, j) \in \{i : i \in S \wedge hop(i) = h_1\}$ 
        $\times \{j : j \in T \wedge hop(j) = h_2\}$  do
9:       if succeed to establish a lightpath from
         node  $i$  to  $j$  then
10:        return path  $\langle s(l), \dots, i, j, \dots, t(l) \rangle$ 
11:      end if
12:    end for
13:  end for
14: end for

```

In Algorithm 3, these edges, with residual bandwidth less than $d(l)$, are removed in line 1 so that the paths in line 4 and 5 computed from the remaining topology satisfy the link capacity constraint. The path for demand l (i.e. $\langle s(l), \dots, t(l) \rangle$) is composed of three parts: path $\langle s(l), \dots, i \rangle$, lightpath $\langle i, j \rangle$, and path $\langle j, \dots, t(l) \rangle$. So, Algorithm 3 finds all nodes, denoted as set S , that have unused interfaces and are connected with node $s(l)$ in line 2. Then, the shortest paths from node $s(l)$ to each node $i \in S$, i.e. $\langle s(l), \dots, i \rangle$, are computed in line 4. We use the hop count as the path cost, and denote the hop count of path $\langle s(l), \dots, i \rangle$ as $hop(i)$. In the same way, Algorithm 3 computes shortest paths $\langle j, \dots, t(l) \rangle$ where $j \in T$ for all nodes (denoted as set T) connected to node $t(l)$ and having unused interfaces in line 3 and 5, and denotes the hop count of path as $hop(j)$. In the end, Algorithm 3 attempts to establish a lightpath $\langle i, j \rangle$ to construct a path for demand l from line 6 to line 14, that is path $\langle s(l), \dots, i, j, \dots, t(l) \rangle$, in the ascending order of hop count. The maximum hop count constraint is considered in line 6 by requiring that the hop count of path $\langle s(l), \dots, i, j, \dots, t(l) \rangle$ (i.e. h) is not more than the maximum hop count h_m . In line 7, h_1 and h_2 represent the hop count of path $\langle s(l), \dots, i \rangle$ and path $\langle j, \dots, t(l) \rangle$ respectively. The hop count from node i to j is one, so we have $h_1 + h_2 = h - 1$. Algorithm 3 tries all combinations of h_1 and h_2 satisfy $h_1 + h_2 = h - 1$ in line 13. In each of this combinations, Algorithm 3 tries all combinations of node $i \in S$ and $j \in T$, where $hop(i) = h_1$ and $hop(j) = h_2$, to establish a lightpath in line 8. If any of these attempts is successful, Algorithm 3 returns the path of demand l and ends.

The complexity of Algorithm 3 is decided mainly by line 9. In the worst case, line 9 need execute $|N| \cdot (|N| - 1)$ times. Since the complexity of each execution of line 9 is

$O(|N|^2)$, the worst-case complexity of Algorithm 3 is $O(|N|^4)$. We implement Algorithm 3 in a more efficient way by constructing set S and T , computing the path $\langle s(l), \dots, i \rangle$ and $\langle j, \dots, t(l) \rangle$ incrementally. In our simulations, we found the general-case complexity of Algorithm 3 is $O(|N|^2)$.

At last, we analyze the complexity of Algorithm 1. The complexity of Algorithm 1 is decided mainly by line 5, 12, 14 and 25. Line 5 and 12 execute $|D1|$ and $|D2|$ times CBR-Dijkstra algorithm respectively, and the complexity of CBR-Dijkstra algorithm is $O(|N|^2)$. Since $|D| = |D1| + |D2|$, the complexity of line 5 and 12 is $O(|D| \cdot |N|^2)$. The worst-case complexity of line 14 is $|L_1| \cdot O(|N|^4)$, and the general-case is $|L_1| \cdot O(|N|^2)$. Let z be the number of executions of line 25, and we have $z \leq |L_1| \leq |L| \leq |D|$. Let $|D_3|_{avg}$ be the average number of traffic flows of each links, and we have $|D_3|_{avg} \leq (|D_2| \cdot |h_m|) / |L|$. So, the complexity of line 25 is $|D_2| \cdot |h_m| \cdot O(|N|^2)$. In conclusion, the worst-case complexity of Algorithm 1 is $O(|D| \cdot |N|^4)$, and the general-case complexity is $O(|D| \cdot |h_m| \cdot |N|^2)$.

5. Performance evaluation

This section evaluates our proposed approach by simulations. The following subsections detail the simulation configuration (i.e., network topologies and the corresponding traffic matrices (TMs)) and parameter settings of the network equipment. The last subsection reports our simulation results with comparisons and analyses.

5.1. Network topologies and traffic matrices

To test the effectiveness of the proposed approaches, we consider two network topologies: the GEANT [31] with 23 nodes and 37 links and the DFN [32] with 58 nodes and 87 links. The GEANT is the pan-European research and education network, and the DFN is the German national research and education network. The choice of the considered network topologies is determined by the availability of the traffic data and their size.

The real traffic matrices for GEANT which include the traffic matrices from January 2005 to April 2005 with a resolution of 15 min are available at [31]. To match the traffic data of 2005 with the current capacity granularity, we scaled all demands by the same factor such that the maximum average traffic demand was 100 Gbps. Then we compute the maximum traffic matrix of all traffic matrices (denoted as TM_m) using the method in [18] which are used to design the base network. We consider the traffic matrices of March 02, 2005, and compute the maximum traffic matrix of each hour as the input of power optimization. As shown in Fig. 1, the total demand of the maximum traffic matrix for each hour during that day is variable greatly with hours. It is worth noting that the total demands in Fig. 1 are only account for 7.12–14.78% of the total demand of TM_m , which confirms that core networks are underutilized severely.

The traffic matrices for DFN are synthetically generated according to the gravity model [33] with the average traffic demand of each node pair of 10 Gbps, 20 Gbps, ..., and

100 Gbps. We compute the maximum traffic matrix of the 10 synthetic traffic matrices as the input of the design of base network, and optimize the power using these 10 traffic matrices to investigate the relationship between the power consumption and the volume of traffic demands.

5.2. Network equipment parameters

The network equipment parameters are set by referring to the data sheets of Cisco CRS-3 Carrier routing system. Cisco CRS-3 provides three types (4, 8, and 16-slot) of single-chassis system with the power of 3080 W, 6160 W, and 12320 W respectively. Cisco CRS-3 multi-chassis routing system [34] is built by interconnecting multiple Line Card Chassis (LCC) which is similar as that of 16-slot single-chassis routing system using one or more Fabric-Card Chassis (FCC) of power 7660 W, and can grow to as many as 72 LCC interconnected by 8 FCC. Specially, the CRS back-to-back multi-chassis system allows two LCCs to be connected directly without the need for a FCC. The nodes of base network can choose the type of routers independently according to the actual need of switch capacity, so the parameter K , $\psi(k)$, $m_l(k)$, and $p_c(k)$ are set as the Table 7 shown.

The line card in Cisco CRS-3 is made up of a Physical Layer Interface Module (PLIM) and a Modular Services Card (MSC). The paper uses Cisco CRS 4-Port 40GE LAN/OTN Interface Module as the PLIM whose power is 185 W, and the power of each MSC is 446 W. The power consumption of network equipment remains almost independent of the actual load [35,4] reported the dynamic power of network switch was less than 15%. We assume the dynamic power occupying 20% of the maximum power by making an optimistic estimation of the technology improvement in recent years. So, the dynamic power for 1 Gbps traffic p_i can be computed, that is 0.70 (W/Gbps). The static power of each interface p_i is 37 W, the static power of each line card is 357 W, and the static power of each type of chassis, as shown in Table 7, can be computed accordingly. The remaining parameters setting are summarized in Table 8.

Table 7
Parameters setting related with the types of chassis.

$k \in K$	$\psi(k)$	$m_l(k)$	$p_c(k)$ (W)
0	1	4	445
1	1	8	890
2	2	16	1780
3	72	16	2460

Table 8
The remaining parameters setting.

Parameters	Values	Parameters	Values
p_r	0.70 W/Gbps	c	40 Gbps
p_i	37 W	μ_m	0.95
p_l	357 W	w	160
m_i	4	τ	0.80
h_m	It depends	α	0.50

5.3. Results and analysis

This section describes our simulation results on the data presented in the previous section. We first present the network power consumption obtained with the CLOD algorithm. Thereafter, the average utilization of network resources (i.e. links, line cards and chassis) is also discussed. At last, we investigate the impact of path length (i.e. hop count) on the network power consumption.

In order to evaluate the performance of the CLOD algorithm, we compare the power consumption computed by the CLOD algorithm with the lower bound of power consumption of model OPT-PO and the power consumption computed by a simple sleeping method which relies on the routing of base network and put the idle network resources into sleep locally by monitoring their utilization. In other words, the simple sleeping method do not change traffic routing of both IP layer and WDM layer, and only put idle interfaces, line cards and chassis into sleep model locally when traffic loads decline. The lower bound value of model OPT-PO can be computed according to (20) and (21). In (20), $n_i(i)_{lb}$ denotes the lower bound of $n_i(i)$ which is the number of active interfaces only enough to transmit and receive the traffic demands of this node. Eq. (21) computes the lower bound of power consumption by using the $n_i(i)_{lb}$ and the corresponding lower bound of $n_i(i)$ and $n_c(i)$ (i.e. $\lceil n_i(i)/m_i \rceil$ and $\lceil n_i(i)/(m_i(i) \cdot m_i) \rceil$), and the lower bound of traffic volume (i.e. $\sum_{l \in D} d(l)$).

$$n_i(i)_{lb} = \left\lceil \max \left(\sum_{l \in D: s(k)=i} d(l), \sum_{l \in D: t(k)=i} d(l) \right) / (\mu \cdot c) \right\rceil \quad \forall i \in N \quad (20)$$

$$LB = \sum_{i \in N} \left(p_c(i) \cdot \lceil n_i(i)_{lb} / (m_i(i) \cdot m_i) \rceil + p_t \cdot \sum_{l \in D} d(l) \right) + p_i \cdot \sum_{l \in D} d(l) \quad (21)$$

5.3.1. Power consumption

In this simulation, we set the parameter h_m with the diameter of the physical network, that is 5 for GEANT and 6 for DFN, to limit the maximum hop count of IP layer. The GEANT network uses the real traffic matrix data which are much lower than the maximum traffic matrix, but the results can reflect the adaptability of the network power consumption to the traffic demand variation during a day. The DFN uses the synthetic traffic matrices, which can be used to investigate the network power consumption of various volumes of traffic demands and the energy-proportional feature of networks.

Fig. 2 depicts the network power consumption of GEANT over a day. In Fig. 2, the adaptation of the network power consumption to the variation of traffic demands is observed compared with Fig. 1. Significant power saving (79.30–86.91%) can be achieved by the CLOD algorithm compared with the highly over-provisioned base network. Moreover, the power consumption obtained with the CLOD algorithm is close to the lower bound of mode OPT-PO, which indicates that the CLOD algorithm can obtain the

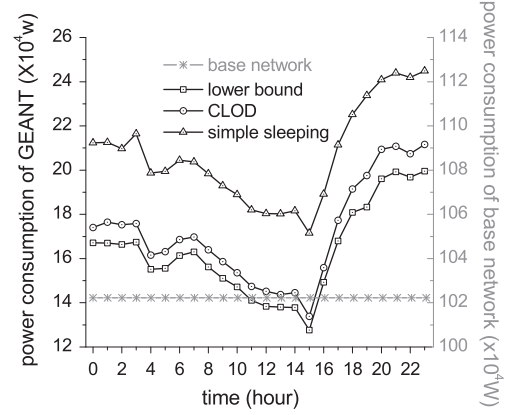


Fig. 2. Power consumption of the GEANT network. The figure shows the network power consumption of GANT over a day computed by the CLOD algorithm compared with the lower bound, and the simple sleeping method. The power consumption of the base network is also presented but with a different scale because its value is much larger than the others.

close to optimal network power consumption. There is also a large gap between the CLOD algorithm and the simple sleeping method, and the CLOD algorithm can save 13.06–21.96% power compared to the simple sleeping method since the CLOD algorithm can reconfigure the virtual topology and change traffic routing to adapt to the dynamic traffic.

Fig. 3 illustrates a near linear relation between the network power consumption achieved by the CLOD algorithm and the average demand, which means that the network is near energy-proportional after the power optimizing of the CLOD algorithm. Similar with Fig. 2, the power savings against the base network are significant. The CLOD algorithm can reduce up to 82.21% power consumption when the traffic demand is low (average demand is 10 Gbps). Note that today's networks are highly underutilized most of time. In Fig. 3, up to 25% power saving can be made by the CLOD algorithm compared to the simple sleeping method, and the gap between them decreases gradually with the growth of average demand. One important cause

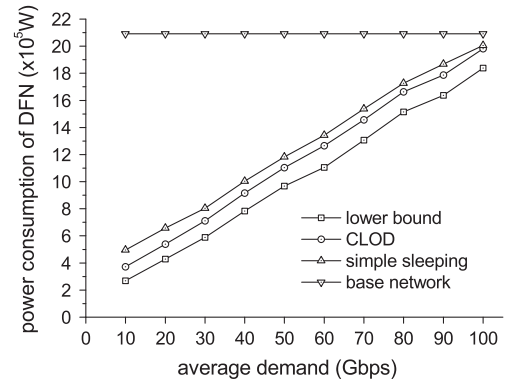


Fig. 3. Power consumption of the DFN network. The figure shows nearly linear relation between the power consumption computed by the CLOD algorithm and the average demand compared with the lower bound, the simple sleeping method and the base network.

is that the traffic grooming used to improve the resource utilization declines gradually with the increasing of traffic demand, which will be discussed further soon afterwards.

5.3.2. Resource utilization

The CLOD algorithm aims to reduce network power consumption by improving network resource utilization and putting idle resources into sleep. There are mainly two factors which can influence resource utilization, which are the traffic grooming and the variation of traffic demands. Traffic grooming can bundle multiple small traffic flows into single big traffic flow by traffic routing to improve resource utilization. Links (or interfaces) and line cards tend to benefit more than chassis from traffic grooming because of their containing relations (i.e. each chassis contains multiple line cards, and each line card contains multiple interfaces). Traffic demands variation can also affect resource utilization. However, the chassis tends to be affected more significant than the interfaces and line cards, because the utilization of interfaces and line cards can be improved by traffic grooming even when the traffic demands are very low.

The average resource utilization in GEANT during a day obtained with the CLOD algorithm is shown as Fig. 4. The average utilizations of links, line cards and chassis change adaptively with the variation of traffic demands. Comparing with links and line cards, chassis experience much greater impact on utilization coming from the traffic demand variation. In contrast, the links and line cards hold much higher average utilization (79.63–92.54%) since links and line cards can benefiting more from the traffic grooming.

Familiar results for DFN network are also shown in Fig. 5. The average utilizations of links, line cards and chassis increase with the growth of traffic demands in general. More significant average utilization increasing with traffic demands (from 42.56% to 81.49%) is observed on chassis. High average utilization (87.81–92.44%) for links and line cards is presented in Fig. 5 because of the traffic grooming, and a slight drop of utilization is also found which may be

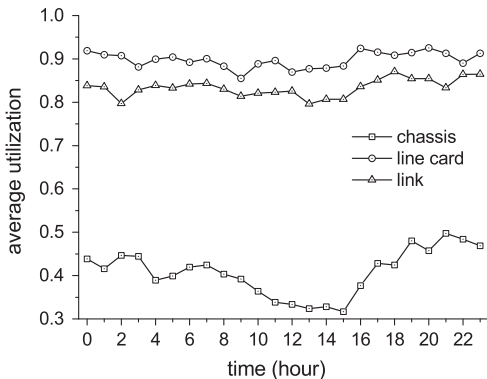


Fig. 4. Average utilizations of links, line cards and chassis in GEANT network over a day obtained with the CLOD algorithm. The links and line cards hold very high average utilizations benefiting from traffic grooming, while the average chassis utilization is influenced more significantly by the traffic demand variation.

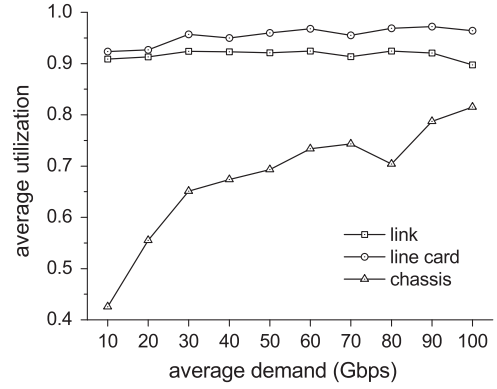


Fig. 5. Average utilizations of links, line cards and chassis in DFN network with various volumes of traffic demand obtained with the CLOD algorithm. The links and line cards hold very high average utilizations benefiting from traffic grooming, while the average chassis utilization is influenced more significantly by the traffic demands.

caused by the decline of traffic grooming. With the increasing of traffic demands, less traffic demands can fill a lightpath, so the needed traffic grooming declines. For example, when the traffic demands are low, 20 traffic flows of 2 Gbps are needed to groom into a lightpath of 40 Gbps. With the increase of traffic demands, the minimum traffic flows is 7 Gbps, so only 5 traffic flows of 7 Gbps are needed to groom into a lightpath of 40 Gbps. In this example, the traffic grooming declines with the increase of the traffic demands, and the maximum utilization of the lightpath decreases from 1 to 0.875.

5.3.3. Impact of hop count on power consumption

The CLOD algorithm can limit the maximum path length (i.e. hop count) of IP layer by parameter h_m , and in the above-mentioned two simulations we set h_m with the diameter of the physical networks. Here, we conduct a series of simulations on GEANT and DFN by setting the parameter h_m with various values from 1 to the diameter of the physical network. We do not consider the values larger than the diameter of the physical network because that may cause overlong network delay, but the network power consumption corresponding to infinity of h_m (i.e. $h_m = \infty$) can be used as reference. Note that $h_m = \infty$ means not limiting the path length. As can be seen in Figs. 6 and 7, as h_m is set to 1, all traffic demands are routed with single-hop paths, so there is no middle nodes in the single-hop path performing traffic grooming to improve the resource utilization, which makes the network power consumption much larger than others. With the increasing of h_m , network power consumption declines firstly and then increases slightly in both GEANT and DFN. More specifically, in Fig. 6, GEANT sees a drastic decrease of network power consumption when h_m increases from 1 to 2, and a slight power consumption drop also is observed with h_m from 2 to 3. When h_m equals 3, 4, and 5, the power consumption curves in Fig. 6 are nearly overlapping, which means the increase of h_m cannot gain additional power saving. Furthermore, the power consumption of GEANT tends to rise with the further growth of h_m , and the

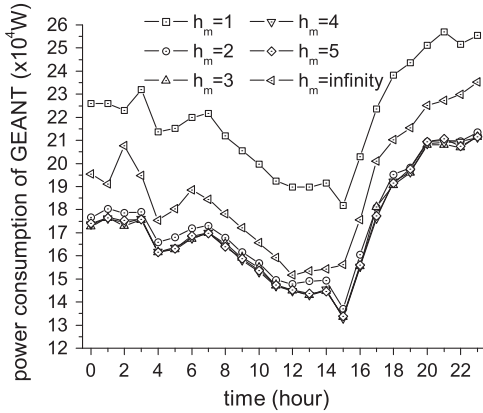


Fig. 6. Power consumption of GEANT network over a day obtained by the CLOD algorithm with various values of h_m .

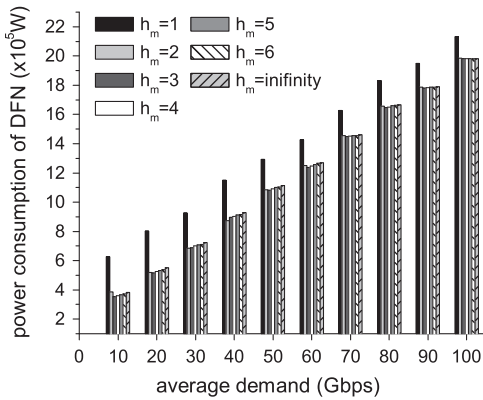


Fig. 7. Power consumption of DFN network with various traffic demands obtained by the CLOD algorithm with various values of h_m .

network power consumption when h_m equals infinity increase 9.96% averagely comparing with h_m equaling 5.

Fig. 7 illustrates the impact of parameter h_m on network power consumption more comprehensively by considering various volumes of traffic demands. The power consumption of DFN declines when h_m increases from 1 to 3, since with the growth of h_m the traffic grooming can improve the utilization of network resources further. However, with the increasing of average traffic demand from 10 Gbps to 100 Gbps, the decreasing amount of network power consumption shrinks gradually because, as aforementioned, traffic grooming declines with the increasing of traffic demands. When the utilization of network resources goes up to a certain extent, increasing h_m will not bring about higher resource utilization, but excessive traffic grooming. As a result, traffic demands are transmitted over longer paths, which results in more traffic volumes (or more dynamic power), and more network resources consumption (or more static power). That is why the network power consumption rises slightly after a drastic drop with the growth of h_m . Due to the decline of traffic grooming, the rising of power consumption is also shrinks with the increasing of traffic demand as shown in Fig. 7.

In conclusion, the increase of h_m can cause network power consumption a drastic drop and then a slight rise. With the growth of traffic demands, the drop and rise both shrinks gradually because of the decline of traffic grooming, and the rise is negligible only when the traffic demand is very high. There exists optimal h_m to minimize network power consumption, and we suggest that h_m can be set by simulations trying the integer between 2 and the diameter of physical network which is small in backbone networks.

6. Conclusions

This work focus on optimizing energy consumption of IP over WDM networks under QoS constraints. We develop formulations to describe the power optimization problem as well as the base network design problem because they are closely related to each other. We show the complexity of the formulations and propose the heuristic CLOD to solve the both problems effectively. CLOD makes use of the skew spatial distribution (i.e. 80–20 law) of traffic demands to construct virtual topology. Different from other heuristics which ignore network resource constraints by assuming ample network resources are provided, CLOD considers resource constraints and can adapt to the available network resources. CLOD also takes account of network performance by controlling the maximum hop count and the maximum link utilization of the whole network. Both the real traffic matrices and synthetic traffic matrices are use to evaluate CLOD. Results indicate CLOD can save power significantly and achieve near energy-proportional IP over WDM networks. The impact of path length on power consumption is also investigated, and we discover that there exists an optimal maximum hop count to minimize network power consumption.

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